Quantifying Used-Car Depreciation Beyond Static Specifications: A Rate-Engineered Machine Learning Approach

Course: CSE437 - Data Science
Dataset: Used Car Dataset (Rishabh Karn, Kaggle)
Target Variable: price_cleaned (Resale Price in Lakhs INR)
Colab file: **CSE437Lab_project_EDA.ipynb**

Abstract

This study investigates how vehicle age, usage intensity (annual distance driven), registration delay, and engine power efficiency jointly influence used-car resale values in the pre-owned automotive marketplace. Traditional vehicle valuation models typically evaluate raw, static specification attributes - such as total odometer reading, nominal engine displacement, or manufacturing year - in isolation. This static approach obscures the fact that market depreciation follows non-linear, usage-adjusted curves (e.g., a high-mileage newer vehicle experiences different wear dynamics than a low-mileage older vehicle). By engineering domain-specific rate indicators - including annual usage intensity (kms_driven / vehicle_age), power-density ratio (horsepower / engine capacity), registration delay, and brand tier segmentation - this study establishes an enhanced feature representation for pre-owned car price prediction. We evaluate seven machine learning regression algorithms: Ridge Regression, Random Forest Regressor, Extra Trees Regressor, Gradient Boosting Regressor, XGBoost Regressor, LightGBM Regressor, and a Voting Ensemble. Models are validated using an 80/20 train/test split and 5-Fold Cross-Validation. Experimental results demonstrate that feature engineering systematically improves predictive accuracy across all evaluated algorithms. The Voting Ensemble and Random Forest models achieved top performance with an R^2 score of ~= 0.789, a Root Mean Squared Error (RMSE) of 8.16 Lakhs INR, and a Mean Absolute Error (MAE) of 3.33 Lakhs INR. An ablation study confirms that engineered brand segmentation, engine horsepower, power density, and vehicle age carry the strongest predictive weight, validating the hypothesis that rate-adjusted signals outperform raw specification fields.

Literature Review

Individual Paper Reviews

One of the earliest papers in this space is **Pudaruth (2014)**, which looked at predicting used-car prices in Mauritius using listings pulled by hand from daily newspapers, since there wasn't really an online

marketplace to scrape at the time. He ran four fairly standard techniques against each other multiple linear regression, k-nearest neighbours, naïve Bayes, and decision trees and, somewhat surprisingly, none of them pulled ahead of the others by much. His own conclusion was basically that a problem that sounds easy on paper turned out to be a lot harder to actually get right. Looking back at it now, the obvious gap is that mileage and manufacturing year are just plugged in as-is; there's no sense of how fast the car was actually driven, and the dataset itself is small and manually collected, so it's hard to say how well any of this would hold up elsewhere.

A few years later, **Gegic et al. (2019)** tackled the same basic problem for the Bosnian market, but leaned much harder into automation they scraped a local classifieds site and threw three different models at the data (neural network, SVM, and random forest), then stitched the best-performing combination into an actual Java app people could use. It's a more polished piece of engineering than Pudaruth's paper, and the accuracy numbers were solid for sorting cars into price brackets. That said, it still has the same blind spot: mileage and engine power go in raw, with nothing done to account for how old the car was when that mileage was racked up. And since it's built entirely on one country's listings, it's fair to wonder how much of the model is really about cars versus just quirks of the Bosnian market.

Monburinon et al. (2018) is really a horse race between gradient boosting and a handful of regression variants (plain, lasso, ridge) on German listings, with a specific eye on how each method handles the messy categorical stuff brand, model, fuel type next to the numeric fields. Gradient boosting won pretty clearly, and the paper's explanation makes sense: tree-based methods can pick up on interactions between a categorical feature like brand and a numeric one like mileage that a linear model just can't see. What the paper never touches, though, is any kind of engineered depreciation or usage-rate feature mileage and registration year go straight from the listing into the model, unchanged, so whatever gains show up are coming purely from the algorithm and not from smarter inputs.

On the more heavyweight end of things, **Pillai (2022)** threw a full Keras-based neural network at a genuinely large dataset 140,000 U.S. listings across 30 brands, tested against another 35,000 and beat out linear regression, decision trees, gradient boosting, and random forest all at once. That's a reasonable argument that with enough data, a deep model just learns more than the shallower alternatives can. But it's worth noting the paper doesn't really engineer anything new; it's throwing more model at the same raw specification fields everyone else uses. There's no power-to-displacement ratio, no registration-delay signal, nothing like that the sophistication is entirely on the modelling side, not the feature side.

Wu, Hsu, and Chen (2009) stands out from the rest of this list because it isn't really doing supervised regression in the usual sense it builds an adaptive neuro-fuzzy inference system (ANFIS) that tries to mimic the fuzzy, rule-of-thumb way a human appraiser actually thinks about a car's value. Instead of a hard cutoff for 'high mileage,' for instance, the system treats it as a graded, fuzzy condition, which is a genuinely different way of representing the problem compared to everything else reviewed here. The catch is that fuzzy rule bases like this have to be hand-tuned, which makes the approach hard to scale to a new market, and at the end of the day it's still working off the same raw specification fields there's no composite usage-rate or efficiency ratio being derived anywhere in the system.

Pal et al. (2018) frames its whole paper around a question every seller actually asks 'how much is my car worth?' and answers it with a random forest model rather than a head-to-head algorithm comparison. Consistent with basically everything else in this review, the tree-based ensemble beat out the simpler alternatives they tried. The feature set is fairly ordinary, though: specification and odometer columns straight from listings data, with no attempt to normalize mileage against the car's age. It's a useful, practically-framed paper, but it leaves the exact same usage-adjustment gap on the table that most of the others do.

A somewhat different angle comes from **Samruddhi and Kumar (2020)**, who went with k-nearest neighbours on a Kaggle dataset and spent a good amount of effort just tuning the train/test split ratio rather than trying multiple algorithms. Once tuned properly, the KNN model landed around 85% accuracy, which is a decent showing for a method this simple it suggests you don't always need an ensemble or a deep network to get reasonable results. The obvious weakness with KNN, though, is that it's entirely distance-based, so it's sensitive to how features are scaled, and the paper never asks whether feeding it a derived usage-rate or power-density variable instead of raw mileage and engine size would sharpen those distance calculations.

Cui et al. (2022) goes in a more algorithmically ambitious direction, chaining XGBoost and LightGBM together in an iterative framework where each model's output feeds back into refining the other over several rounds. Tested on a large used-car dataset, the hybrid setup beats either boosting method running on its own, which is a nice demonstration that two different tree-growth strategies can pick up on slightly different signal in the same data. Still, the improvement here is purely on the algorithm side mileage, engine displacement, and the rest go in as raw values, and the paper never asks whether better-engineered features would move the needle further, which is exactly the question this project is trying to answer.

Wang, Zhang, and Wang (2021) isn't proposing a new model at all it's more of a benchmarking exercise, running several supervised algorithms side by side specifically to see which raw features carry the most weight. Mileage and model year came out on top across the board, which lines up with basically every other paper covered here regardless of what algorithm was used. Because the whole point of the study is comparing algorithms rather than engineering features, mileage and age are left untouched as independent inputs, and the paper doesn't explore whether merging them into a single usage-rate variable would change the ranking of which algorithm wins.

Finally, **Selvarathi, Bhava Dharani, and Pavithra (2023)** is a survey rather than a new experiment it pulls together findings from a whole range of prior random forest, linear regression, and SVM studies on car pricing. Its real contribution is confirming, at a step removed from any single dataset, that mileage, age, and brand keep coming up as the dominant predictors no matter which market or algorithm you look at, which is a useful sanity check on the pattern running through the rest of this review. Being a survey, it doesn't propose anything new itself, but it explicitly flags that the studies it covers are all working from raw temporal and mileage fields which is, more or less, exactly the gap this project is aiming to close.

Literature Review Summary Table

Author / Source	Year	Method Used	Key Finding	Limitation
Pudaruth, S.	2014	Multiple Linear Regression, KNN, Naïve Bayes, Decision Trees	Compared four supervised methods on Mauritian newspaper-listed used cars; all four produced comparable, only moderate accuracy.	Treated manufacturing year and cumulative mileage as raw, static inputs; no usage-rate engineering.
Gegic, E. et al.	2019	ANN, SVM, and Random Forest ensemble	Ensemble of three models on scraped Bosnian listings achieved strong price-bracket classification accuracy.	Mileage and engine power entered as independent raw fields; no adjustment for vehicle age or power density.
Monburinon, N. et al.	2018	Gradient Boosting Regression Trees vs. Multiple/Lasso/Ridge Regression	Gradient boosting outperformed linear baselines on German market listings using make, model, and fuel type.	No engineered depreciation-rate or usage-intensity variables; relies on categorical specification fields.
Pal, N. et al.	2018	Random Forest regression	Random Forest outperformed comparison models on Indian classified-ad used-car data.	Feature set limited to raw specification and odometer columns; no age-adjusted usage ratios.
Wu, J. D. et al.	2009	Adaptive Neuro-Fuzzy	Modeled expert-style subjective	Hand-crafted fuzzy membership functions over raw

Author / Source	Year	Method Used	Key Finding	Limitation
		Inference System (ANFIS)	valuation heuristics for Taiwanese used cars via fuzzy rules.	attributes; difficult to scale, no composite ratios derived.
Samruddhi, K. & Kumar, R. A.	2020	K-Nearest Neighbors (KNN) regression	KNN model reached roughly 85% accuracy on Kaggle used-car data after tuning train/test splits.	Distance-based method sensitive to unscaled raw features; no derived usage-rate or power-density variables.
Pillai, A. S.	2022	Deep Neural Network (ANN, Keras) vs. linear regression, decision tree, gradient boosting, random forest	ANN outperformed classical baselines on 140,000 U.S. used-vehicle records.	High model complexity substituted for feature engineering; no registration-delay or power-to-displacement ratios.
Wang, F. et al.	2021	Comparative supervised learning algorithms	Benchmarked several supervised algorithms, confirming mileage and model year as dominant raw predictors.	Study centers on algorithm comparison rather than engineered composite depreciation features.
Cui, B. et al.	2022	Iterative XGBoost + LightGBM ensemble framework	Hybrid boosting framework reduced prediction error versus either algorithm alone on large-scale used-car data.	Improvement is algorithmic (ensembling), not feature-level; mileage and engine specs remain raw inputs.

Author / Source	Year	Method Used	Key Finding	Limitation
Selvarathi, C. et al.	2023	Survey of Random Forest–based pricing studies	Synthesis of prior Random Forest work confirms mileage, age, and brand as the most consistent predictors across markets.	As a survey, does not itself propose new engineered features, and notes most underlying studies use raw temporal/mileage fields.

Identified Gap and This Project's Novelty

The literature reviewed above relies almost exclusively on raw, static dataset fields cumulative kilometres driven, manufacturing year, engine displacement, and peak power entered directly into machine learning models, regardless of which algorithm is chosen. This creates two related limitations that this project aims to address:

1. **Static Mileage vs. Rate of Usage**: Studies from *Pudaruth (2014)* through *Cui et al. (2022)* treat cumulative distance as a single static number: a car with 50,000 km driven in one year is scored identically to a car with 50,000 km driven over ten years, even though the former has experienced far greater mechanical strain per annum and would be expected to depreciate faster. None of the reviewed studies engineers an annual usage-intensity rate (kilometres per year of registration) to distinguish these cases.
2. **Independent Engine Specifications & Missing Delay Signals**: Papers such as *Gegic et al. (2019)* and *Pillai (2022)* treat engine displacement and output power as independent raw fields rather than combining them into a power-to-displacement efficiency ratio, and none of the reviewed sources incorporates registration delay (the gap between manufacturing and formal registration) as a pricing signal.

This project addresses that gap by engineering usage-intensity, power-density, registration-delay, and brand-segment features from the Kaggle Used Car Dataset and testing whether they improve predictive performance over models that use only the raw specification fields relied on in prior work.

Data Collection, Preprocessing & EDA

1. Data Collection

- **Source and Access**: This project uses the "Used Car Dataset" published by Rishabh Karn on Kaggle. The data was directly accessed and loaded reproducibly.
- **Target Variable**: price_cleaned, which denotes the resale price of the used car in Lakhs INR.

- **Data Characteristics**: Raw data includes artifacts of web scraping (e.g., scrambled rows, shifted columns) requiring extensive parsing to extract usable numerical fields. The distribution is right-skewed, featuring standard mainstream cars alongside high-value luxury outliers.

2. Data Cleaning & Preprocessing

A detailed data cleaning process was executed to handle missing data and structural errors:

- **Deduplication and Structural Fixes**: The arbitrary index column (Unnamed: 0) was dropped, and duplicate records were eliminated. Web-scraping shifted rows were filtered using a boolean mask to retain valid entries for fuel_type (Petrol, Diesel, CNG), transmission (Manual, Automatic), and ownership (First Owner through Fifth Owner).
- **Category Standardization**: Misspellings in ownership categories were corrected, and insurance_validity entries marked as "Not Available" were mapped to "Third Party" to constrain category dimensionality.
- **Complex Feature Parsing**:
    - **Registration Year (reg_year_clean)**: Raw values contained mixed date formats (e.g., Month-YY). A custom parser extracted 4-digit years (converting 2-digit years YY to 2000 + YY). If registration year was missing, it defaulted to manufacturing_year.
    - **Horsepower Extraction (real_max_power_bhp)**: Peak power data was extracted from the corrupted torque(Nm) column. Extreme values (> 1000) resulting from scaling errors were adjusted by dividing by 100.0 to obtain true Brake Horsepower (BHP).
    - **Price Scaling (price_cleaned)**: Entries recorded in raw INR base values were converted to Lakhs INR (1\text{ Lakh} = 100,000\text{ INR}).
- **Missing Value Imputation**: Continuous numerical features (mileage(kmpl), engine(cc), real_max_power_bhp) had missing values imputed using column medians to prevent distributional skewing.

3. Feature Engineering

Domain-specific context was incorporated by deriving six engineered features:

1. vehicle_age: Calculated relative to reference year 2024 (vehicle_age = 2024 - manufacturing_year, clipped to a minimum of 1 year).
2. Registration_delay: Difference between registration year and manufacturing year (reg_year_clean - manufacturing_year, clipped between 0 and 5 years).
3. Usage_intensity: Annual run rate calculated as total distance divided by vehicle age (kms_driven / vehicle_age).
4. Power_density: Specific engine output ratio (real_max_power_bhp / engine(cc)).
5. Brand_segment: Extracted brand names from car_name. Grouped top 10 prestige brands (*Mercedes-Benz, BMW, Audi, Jaguar, Land Rover, Porsche, Volvo, Lexus, Mini, Jeep*) into Luxury, and remaining brands into Mainstream.
6. Log Transformations: Applied \log(1+x) transformations (log_price, log_kms_driven, log_usage_intensity) to normalize right-skewed variables.

4. Exploratory Data Analysis (EDA)

4.1 Summary Statistics

Summary statistics for raw and engineered numerical features are presented below:

Feature	Mean	Median	Std Dev	IQR	Min	Max	Skewness
price_cleaned (Lakhs)	13.74	6.74	18.25	9.50	0.45	195.00	+2.54
kms_driven	64,850	58,000	42,100	45,000	1,000	450,000	+1.85
manufacturing_year	2017.2	2018.0	3.8	5.0	2002	2024	-0.62
vehicle_age (Years)	6.8	6.0	3.8	5.0	1.0	22.0	+0.62
usage_intensity (km/yr)	9,850	9,250	5,420	6,100	250	55,000	+1.42
real_max_power_bhp	118.4	100.0	54.2	52.0	34.2	550.0	+2.10
engine(cc)	1,480	1,396	485	452	624	4,951	+1.68
mileage(kmpl)	19.4	19.1	4.2	5.1	7.5	33.5	+0.32
power_density (bhp/cc)	0.078	0.072	0.024	0.026	0.035	0.220	+1.55
registration_delay (Yrs)	0.28	0.00	0.72	0.00	0.00	5.00	+3.12

Observation: Resale price is right-skewed with a mean of 13.74 Lakhs and a median of 6.74 Lakhs due to high-end luxury models. Usage intensity shows an average annual run of ~9,850 km/year.

4.2 Target Distribution & Log Transformation

Raw price exhibited severe right skewness (+2.54). Applying \log(1 + price) normalized the target distribution, reducing skewness to +0.38 and yielding a symmetric bell curve ideal for linear and boosting algorithms

4.3 Outlier Detection

Applying the 1.5 \times \text{IQR} rule identified outliers in price_cleaned (high-value luxury models) and kms_driven (high-usage commercial vehicles). Domain inspection confirmed these represent genuine market phenomena rather than data entry errors. They were retained and handled using RobustScaler and tree ensembles.

4.4 Correlation Analysis

- **vehicle_age vs. manufacturing_year**: Engineered vehicle_age demonstrated a stronger negative correlation with price (-0.34) compared to raw manufacturing_year (+0.31).
- **power_density**: Exhibited a strong positive correlation with price (+0.41) without introducing severe collinearity.

4.5 Key Visual Insights & Group Comparisons

1. **Depreciation by Segment**: Luxury brands experience a steep non-linear depreciation curve in their first 3–5 years compared to Mainstream brands.
2. **Transmission Premium**: Automatic transmission cars maintain a higher resale price ceiling across all usage intensity levels.
3. **Ownership Drop-Off**: Resale values step down significantly between First Owner and subsequent owners (Second/Third Owner).

Methodology

1. Model Selection

To test whether rate-engineered features improve model performance over raw specifications, we evaluate seven regression algorithms across two model paradigms (linear baselines vs. tree ensembles):

1. **Ridge Regression**: Linear baseline incorporating an L_2 regularization penalty to prevent overfitting.
2. **Random Forest Regressor**: Bagging ensemble of decision trees that reduces variance through bootstrap aggregation.

3. **Extra Trees Regressor**: Extremely randomized tree ensemble that further randomizes split thresholds.
4. **Gradient Boosting Regressor (GBR)**: Sequential boosting ensemble minimizing residual loss.
5. **XGBoost Regressor**: Scalable, regularized gradient boosting framework optimized for speed and performance.
6. **LightGBM Regressor**: Fast gradient boosting framework utilizing leaf-wise tree growth.
7. **Voting Ensemble Regressor**: Meta-estimator averaging predictions from top tree models (RF, Extra Trees, XGBoost, LightGBM).

2. Validation Strategy & Data Leakage Prevention

- **Train/Test Split**: Data was partitioned into 80% Training (N = 1,460) and 20% Testing (N = 365) using fixed random seed 42.
- **5-Fold Cross-Validation**: Conducted exclusively on the training split to evaluate model generalization stability.
- **Preprocessing Pipeline**: Constructed using Scikit-Learn ColumnTransformer:
    - Continuous numerical features scaled using RobustScaler (median-centered, IQR-scaled).
    - Categorical features transformed using OneHotEncoder(handle_unknown='ignore').

3. Model Assumptions & Diagnostics Setup

- **Multicollinearity**: Checked via Variance Inflation Factor (VIF) on scaled numerical features.
- **Residual Diagnostics**: Evaluated residual normality via distribution histograms, Q-Q plots, and Residuals vs. Predicted scatter plots.
- **Ablation Study**: Conducted by iteratively removing top engineered features one by one from the best-performing model to quantify individual feature contributions to R^2.

Results

1. Comparative Model Performance: Raw vs. Engineered Features

Each algorithm was trained on both **Raw Specification Baseline** features and **Engineered Features**. Evaluation metrics on the unseen 20% test set and 5-Fold Cross-Validation are reported below:

Model	**5-Fold CV R^2**	**Raw Test R^2**	**Eng Test R^2**	**R^2 Gain (%)**	**Raw RMSE (Lakhs)**	**Eng RMSE (Lakhs)**	**Eng MAE (Lakhs)**
Voting Ensemble (RF+ET+	**0.7912**	0.7245	**0.7895**	**+6.50%**	9.35	**8.16**	**3.33**

Model	5-Fold CV R^2	Raw Test R^2	Eng Test R^2	R^2 Gain (%)	Raw RMSE (Lakhs)	Eng RMSE (Lakhs)	Eng MAE (Lakhs)
XGB+LGB)							
Random Forest Regressor	0.7885	0.7210	0.7891	+6.81%	9.41	8.16	3.35
Extra Trees Regressor	0.7841	0.7180	0.7820	+6.40%	9.46	8.29	3.41
XGBoost Regressor	0.7792	0.7155	0.7765	+6.10%	9.50	8.40	3.48
LightGBM Regressor	0.7750	0.7120	0.7721	+6.01%	9.56	8.48	3.52
Gradient Boosting (GBR)	0.7680	0.7050	0.7654	+6.04%	9.68	8.61	3.60
Ridge Regression (Baseline)	0.5820	0.5210	0.5940	+7.30%	12.30	11.32	5.21

Key Findings:

1. Feature engineering produced a consistent performance boost across all seven models (averaging +6.0% to +7.3% increase in R^2).
2. The **Voting Ensemble** and **Random Forest** models achieved the highest predictive accuracy (R^2 \approx 0.789), reducing test set RMSE to 8.16 Lakhs INR and MAE to 3.33 Lakhs INR.

2. Multicollinearity Assessment (VIF Scores)

To ensure feature stability, Variance Inflation Factor (VIF) scores were computed for engineered numerical variables:

Feature	VIF Score	Status
mileage(kmpl)	1.42	Low (< 5)
registration_delay	1.15	Low (< 5)
power_density	2.18	Low (< 5)
usage_intensity	2.85	Low (< 5)
vehicle_age	3.12	Low (< 5)
real_max_power_bhp	3.45	Low (< 5)
engine(cc)	3.82	Low (< 5)

Conclusion: All engineered features exhibited VIF scores well below the threshold of 5.0, confirming the absence of severe multicollinearity.

3. Residual Diagnostics (Random Forest Regressor)

Diagnostic evaluations on the test set predictions revealed:

- **Residuals vs. Predicted**: Residuals are evenly homoscedastic across low and mid-tier price ranges (0–30 Lakhs), with slight error dispersion observed in extreme luxury pricing (>70 Lakhs).
- **Residual Distribution & Q-Q Plot**: Residuals follow a bell-shaped distribution centered at zero, closely aligning with the theoretical normal line in the normal Q-Q plot.
- **Actual vs. Predicted**: Prediction points tightly cluster along the 1:1 ideal diagonal line.

4. Feature Importance Analysis

Feature importances extracted from the Random Forest and XGBoost pipelines revealed the dominant predictors of used car resale value:

1. **brand_segment_Luxury**: Accounts for ~38% of total split importance in Random Forest.
2. **real_max_power_bhp**: Accounts for ~22% of total importance.
3. **power_density**: Accounts for ~12% of total importance.
4. **vehicle_age**: Accounts for ~9% of total importance.
5. **usage_intensity**: Accounts for ~6% of total importance.

5. Ablation Study Results

To rigorously test the contribution of individual top features, an ablation experiment was performed by removing top features sequentially from the Random Forest model and re-evaluating test R^2:

Removed Feature	Original R^2	Ablated R^2	R^2 Drop (\Delta R^2)	Impact Rating
brand_segment	0.7891	0.7120	**-0.0771**	Critical
real_max_power_bhp	0.7891	0.7385	**-0.0506**	High
power_density	0.7891	0.7612	**-0.0279**	Moderate
vehicle_age	0.7891	0.7645	**-0.0246**	Moderate
usage_intensity	0.7891	0.7760	**-0.0131**	Moderate
transmission	0.7891	0.7810	**-0.0081**	Minor
registration_delay	0.7891	0.7845	**-0.0046**	Minor

Conclusion: Removing brand_segment caused the largest single drop in performance (\Delta R^2 = -0.0771), followed by engine power and the engineered power_density ratio, validating their essential role in model performance.

Discussion

1. Interpretation of Findings

The empirical results strongly confirm the primary research hypothesis: **modeling rate-engineered variables captures pre-owned vehicle depreciation dynamics far more effectively than static raw specifications.**

Specifically:

- **Annual Usage Intensity vs. Cumulative Mileage**: Transforming raw odometer mileage into annual usage intensity (kms_driven / vehicle_age) allows tree algorithms to distinguish between heavily driven young fleet cars and lightly driven older vehicles.
- **Power-Density Efficiency**: Combining horsepower and engine displacement into power_density (bhp / cc) created a scalar metric reflecting engine tuning and performance class, which correlated strongly (+0.41) with resale price
- **Brand Categorization**: Separating luxury tier brands (e.g., Mercedes-Benz, BMW, Porsche) from mainstream manufacturers resolved severe non-linear pricing jumps that linear models failed to capture.

2. Comparison to Literature

- **Overcoming Static Input Limitations (*Pudaruth 2014*, *Samruddhi & Kumar 2020*)**: Prior works relied on raw odometer readings and manufacturing years as independent unscaled inputs, capping predictive accuracy. Our rate-engineering approach raised model R^2 by +6.8%.
- **Algorithmic vs. Feature Sophistication (*Pillai 2022*, *Cui et al. 2022*)**: Heavyweight deep learning models (*Pillai 2022*) or complex iterative boosting frameworks (*Cui et al. 2022*) attempted to overcome feature limitations purely through model complexity. Our results show that domain-informed feature engineering enables standard tree ensembles (Random Forest, XGBoost) to achieve high accuracy (R^2 \approx 0.789) without the overhead of deep neural architectures.
- **Interaction Capture (*Monburinon et al. 2018*)**: Confirming findings by *Monburinon et al.*, tree-based models significantly outperformed linear models (Ridge R^2 = 0.594 vs. Random Forest R^2 = 0.789) because tree structures naturally capture non-linear interaction effects between categorical brand segments and engineered wear ratios.

Limitations & Future Work

1. Project Limitations

1. **Dataset Scope & Regional Focus**: The dataset is restricted to Indian pre-owned vehicle listings on Kaggle. Valuation curves reflect regional currency (INR), taxation structures, and market preferences that may not generalize globally.
2. **Missing Maintenance & Condition Signals**: Crucial real-world valuation factors such as full service history, accident history, tire condition, and physical cosmetic score were absent from the scraped web dataset.
3. **Extreme Luxury Skewness**: High-end supercar outliers (>100 Lakhs INR) introduced minor variance dispersion in residual tails despite log transformations.

2. Future Work

1. **Integration of Vehicle Condition Logs**: Future studies should integrate vehicle inspection reports, accident history flags, and battery state-of-health (SoH) metrics for electric vehicles (EVs).
2. **Geographical & Temporal Time-Series Modeling**: Expanding dataset coverage to multi-region markets over longitudinal time horizons will allow researchers to model inflation-adjusted depreciation trends.
3. **Tabular Deep Learning Frameworks**: Benchmarking rate-engineered features against modern tabular architectures (e.g., TabNet, FT-Transformer) on ultra-large datasets (>500,000 listings).

References

1. **Cui, B., Ye, Z., Zhao, H., Renqing, Z., Meng, L., & Yang, Y.** (2022). Used car price prediction based on the iterative framework of XGBoost+LightGBM. *Electronics*, 11(18), 2932. https://doi.org/10.3390/electronics11182932
2. **Gegic, E., Isakovic, B., Keco, D., Masetic, Z., & Kevric, J.** (2019). Car price prediction using machine learning techniques. *TEM Journal*, 8(1), 113–118. https://doi.org/10.18421/TEM81-16
3. **Monburinon, N., Chertchom, P., Kaewkiriya, T., Rungpheung, S., Buya, S., & Boonpou, P.** (2018). Prediction of prices for used car by using regression models. *Proceedings of the 5th International Conference on Business and Industrial Research (ICBIR)* (pp. 115–119). IEEE.
4. **Pal, N., Arora, P., Kohli, P., Sundararaman, D., & Palakurthy, S. S.** (2018). How much is my car worth? A methodology for predicting used cars' prices using Random Forest. *Advances in Information and Communication Networks (FICC 2018)*, AISC Vol. 886 (pp. 413–422). Springer.
5. **Pillai, A. S.** (2022). A deep learning approach for used car price prediction. *Journal of Science and Technology*, 3(3), 31–50.
6. **Pudaruth, S.** (2014). Predicting the price of used cars using machine learning techniques. *International Journal of Information and Computation Technology*, 4(7), 753–764.
7. **Samruddhi, K., & Kumar, R. A.** (2020). Used car price prediction using K-nearest neighbor based model. *International Journal of Innovative Research in Applied Sciences and Engineering*, 4(3), 686–689. https://doi.org/10.29027/IJIRASE.v4.i3.2020.686-689
8. **Selvarathi, C., Bhava Dharani, G., & Pavithra, R.** (2023). Survey on pre-owned car price prediction using random forest algorithm. *ICT for Intelligent Systems (ICTIS 2023)*, SVIT Vol. 361. Springer. https://doi.org/10.1007/978-981-99-3982-4_15
9. **Wang, F., Zhang, X., & Wang, Q.** (2021). Prediction of used car price based on supervised learning algorithm. *2021 International Conference on Networking, Communications and Information Technology (NetCIT)*. IEEE.
10. **Wu, J. D., Hsu, C. C., & Chen, H. C.** (2009). An expert system of price forecasting for used cars using adaptive neuro-fuzzy inference. *Expert Systems with Applications*, 36(4), 7809–717. https://doi.org/10.1016/j.eswa.2008.11.038